GenePath

From one pharmacogenomic report to two real decisions · DSH Hacks V2 · AI × Healthcare

Live demo: candicewu0515.github.io/genepath · **Code:** github.com/candicewu0515/genepath

PROBLEM

Pharmacogenomic tests are now cheap and common, but the report a patient receives says things like “CYP2C19 *2/*2”. Most people cannot tell whether that matters for a medicine they already take. These variants are also far more common in some ancestries than others, so a one-size-fits-all dose quietly fails specific populations.

SOLUTION

GenePath is a single web page that answers the two questions patients actually ask:

- **Is my medication affected?** Pick one of 16 medicines across 8 genes — blood thinners, statins, acid-reflux drugs, antidepressants, painkillers, chemotherapy, immune suppressants — and a genotype. A printable card explains the result in plain words, quotes the CPIC guideline verbatim, charts the variant across six ancestry groups, and lists questions for the next appointment.
- **Is there a trial that fits me?** Pick one of 9 conditions plus age, sex and city. GenePath queries ClinicalTrials.gov live, widens the search if nothing is nearby, keeps studies open to that person, and explains each study’s key entry rules in plain words, with contact details.

HOW AI IS USED

- The app and its data pipeline were built with AI coding assistants in under two days.
- The pipeline pulls guideline and frequency data from two public APIs and fails the build if any value cannot be confirmed, so no clinical text is typed by hand.
- At run time the page makes **no AI calls**, so nothing a user enters reaches a model and the output cannot hallucinate.

DATA (ALL PUBLIC, ALL REAL)

- **All of Us** public Data Browser API: allele frequencies for 11 variants from ~535,000 participants, re-verified by script.
- **CPIC API**: diplotype, phenotype and recommendation text, verbatim, for 16 drugs across 8 genes. Warfarin alone follows the FDA label table, as it is dosed by algorithm.
- **ClinicalTrials.gov API v2**: live browser query, with a saved 9-condition snapshot as offline fallback.

Allele frequency	African	European	East Asian	South Asian
CYP2C19*2 (clopidogrel, PPIs, SSRIs)	17.9%	14.6%	30.8%	34.4%
VKORC1 -1639A (warfarin)	9.4%	38.6%	87.4%	17.1%
CYP2C9*2 (warfarin, NSAIDs)	2.1%	12.9%	0.1%	4.5%
SLCO1B1*5 (statins)	2.8%	16.0%	12.7%	4.8%
NUDT15*3 (azathioprine)	0.1%	0.3%	9.6%	6.5%
DPYD*2A (5-FU, capecitabine)	0.1%	0.5%	0.0%	0.5%

PRIVACY AND SAFETY BY DESIGN

- Static page: no account, no server, no storage. The only network request is the public trial search (condition and city).
- Guideline information is framed for a conversation with a clinician, never as an instruction; no doses are recommended.

LIMITATIONS

- Common variants only (CYP2C9 *5/*6/*8/*11 and CYP2D6, which needs copy-number calling, are not covered). Where a guideline uses two genes, the card fixes the second at normal function and says so.
- Frequencies are population-level allele frequencies, not individual risk; only a study team can confirm trial eligibility.

NEXT STEPS

Upload of a real PGx report file, CYP2D6 support (codeine, tramadol, antidepressants), genotype-aware trial matching, and Spanish and Chinese versions.

Disclaimer. GenePath is an educational demonstration and does not provide medical advice. Do not start, stop or change any medication based on it.